

Title

Impact of electronic nicotine devices combined with counseling on sleep outcomes in tobacco-dependent smokers: secondary analysis of a pragmatic cessation trial

Abstract

Background: While electronic nicotine delivery devices can support smoking cessation, their effects on sleep quality during cessation attempts are not well characterized.

Methods: This secondary analysis evaluated sleep quality in 831 adult smokers from the ESTxENDS trial conducted across five Swiss sites. Participants were randomized to receive either free electronic nicotine devices plus standard smoking-cessation counseling (n=~415) or counseling alone (n=~416) for six months. Sleep quality was assessed using a validated standardized sleep questionnaire at baseline and six-month follow-up. Component measures of sleep quality including efficiency, duration, latency, and disturbance were examined. **Results:** Mean sleep quality scores did not differ significantly between the electronic nicotine device plus counseling group and counseling-alone group at six months (adjusted between-group difference: -0.20 points, 95% CI -0.53 to 0.13; p=0.24). Among subscale analyses, only sleep efficiency showed statistically significant improvement in the electronic nicotine device arm (adjusted between-group difference 1.87 points; p=0.02); however, this difference was below the threshold considered clinically meaningful. No differences were observed for sleep onset time, sleep duration, or nighttime disturbance measures. **Conclusions:** In smokers attempting cessation with either approach, electronic nicotine device use adjunctive to counseling did not substantially alter overall sleep quality compared to counseling alone. Clinicians can inform patients attempting smoking cessation via electronic devices that this approach is unlikely to adversely affect their sleep.